

F O R T I M A N A G E R N S E 6

Lab 07 — Policy Packages, Objects and Per-Device Mapping

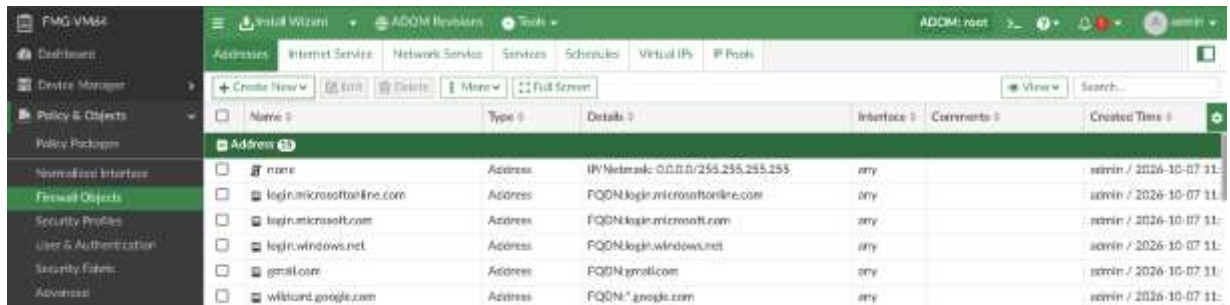
01. Objects in an ADOM

Labs 05 and 06 worked on device-level configuration — interfaces, DNS, routes — all of it scoped to one device database. Policy is different. Firewall policies and the objects they reference live in the ADOM, shared across every device in it, and are pushed as a policy package rather than as device settings.

The lab continues on the pair from Lab 06: FortiManager 7.4.3 and FortiGate 7.2.1 (fortigate7-2, 10.58.228.177) in the root ADOM.

► 1.1 The object database

Policy & Objects > Firewall Objects holds the addresses, services, schedules and groups available to every policy package in the ADOM. Objects are defined once here and referenced by name wherever they are used.



Name	Type	Details	Interface	Comments	Created Time
none	Address	IP Netmask: 0.0.0.0/255.255.255.255	any		admin / 2026-10-07 11:11
login.microsoftonline.com	Address	FQDN:login.microsoftonline.com	any		admin / 2026-10-07 11:11
login.microsoft.com	Address	FQDN:login.microsoft.com	any		admin / 2026-10-07 11:11
login.windows.net	Address	FQDN:login.windows.net	any		admin / 2026-10-07 11:11
gmail.com	Address	FQDN:gmail.com	any		admin / 2026-10-07 11:11
wikipedia.google.com	Address	FQDN:wikipedia.google.com	any		admin / 2026-10-07 11:11

Figure 1 — Firewall Objects — the ADOM-wide address list.

► 1.2 Creating an address

An address object named LAN-Net was created for the site network.



Create New Address

Category:
 Name: LAN-Net
 Color:
 Type: Subnet
 IP/Netmask: 192.168.10.0/0.0.0.0
 Interface: any
 Static Route Configuration:
 Comments:
 Add To Groups:
 Click to select

Figure 2 — Create New Address — LAN-Net, type Subnet.

Observation	The IP/Netmask field was filled in as 192.168.10.0/0.0.0.0 — the address is right, the mask is zero.
Effect	A zero mask makes the object 0.0.0.0/0. FortiManager accepted it without complaint, and the object is now functionally identical to the built-in all.
Consequence	This single typo is what produces the failed install in section 03 and the Conflict state in section 04. It is worth following through the lab rather than correcting quietly.

► 1.3 Change notes

Saving the object prompts for a change note before it is committed. The field is mandatory, which is a workspace behaviour rather than an object behaviour — the note is attached to the ADOM revision the save creates.



Figure 3 — The mandatory change note prompt.

These notes are the ADOM-level equivalent of the install comments in Lab 05. They are the only place a reason for a change is recorded, and they are what a later revision diff is read against.

► 1.4 Creating a service

A custom service was created for OPC UA, the protocol this policy is meant to govern. OPC UA binary runs on TCP 4840, which is not in the FortiGate's predefined service list.

The screenshot shows the 'Create New Service' dialog box. The 'Name' field contains 'OPC-UA'. The 'Comments' field is empty. The 'Color' field has a 'Change' button. The 'Service Type' is set to 'Firewall' (with 'Proxy' as an alternative). The 'Category' is 'Click to select'. The 'Protocol Type' is 'TCP/UDP/SCTP'. The 'IP/FQDN' is '0.0.0.0'. The 'Add To Groups' field is empty. Below these fields is a table with columns: Protocol, Source Port, Destination Port, and Action. The table contains one row: TCP, 1, 65535, 4840, 4840. The 'Advanced Options' and 'Revision' sections are collapsed.

Protocol	Source Port	Destination Port	Action
TCP	1	65535	4840

Figure 4 — Create New Service — OPC-UA on TCP destination port 4840.

The source port range is left at the default 1 to 65535. Narrowing the source port on a client-initiated protocol is a common mistake — clients use ephemeral ports, so restricting the source range breaks the service rather than tightening it.

02. Policy Packages

A policy package is an ordered set of firewall policies that is installed to one or more devices as a unit. The package is edited in FortiManager and exists independently of any device until it is assigned an installation target.

► 2.1 Creating a policy

The screenshot shows the 'Create New Firewall Policy' dialog box. The policy name is 'OPC-UA-POLICY'. The type is 'Standard'. The incoming and outgoing interfaces are both 'any'. The source is 'LAN-Net' (0.0.0.0/0.0.0.0). The destination is 'all' (0.0.0.0/0.0.0.0). The service is 'OPC-UA' (TCP/4840). The 'Negate Source' and 'Negate Destination' options are disabled. The 'IP/MAC Based Access Control' and 'Logical And With Secondary Tags' options are also disabled.

Figure 5 — Create New Firewall Policy — OPC-UA-POLICY using the LAN-Net and OPC-UA objects.

Note what the Source field displays beneath the object name: IP/Netmask: 0.0.0.0/0.0.0.0. The policy editor is showing the resolved value of LAN-Net, and it already reveals the problem introduced in section 1.2 — the object the policy sources from matches every address.

Observation	The policy Action is DENY, both in the editor and in the installed policy on the device.
Why	Deny is FortiManager's default action for a new policy. Nothing in the workflow prompts for it to be changed.
Check	A policy named OPC-UA-POLICY that denies OPC UA is almost certainly not the intent. Worth confirming before the package is reused as a baseline.

☐	#	Name	From	To	Source	Destination	Schedule	
☐	1	OPC-UA-POLICY	☐ any	☐ any	LAN-Net	all	always	

Figure 6 — The policy present in the package.

► 2.2 Objects are reference-counted

Deleting an object that a policy uses is refused — or rather, it is allowed but only after FortiManager has said what will break.

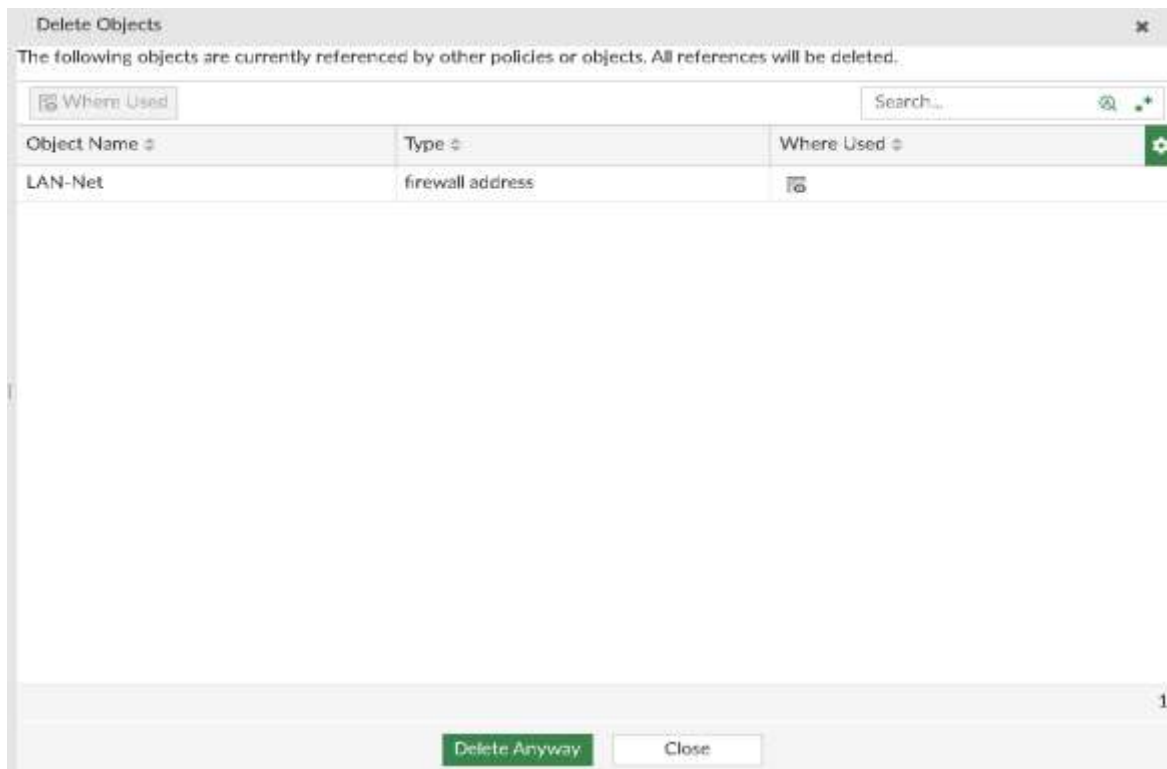


Figure 7 — Delete Objects — LAN-Net reported as referenced, with a Where Used link.

Where Used is the dependency map

FortiManager tracks every reference to every object, which is what makes shared objects safe to use across packages.

The Delete Anyway button removes the object and every reference to it, including from policies that are currently installed. The reference count in the object list is the quick version of the same information.

03. Installing a Policy Package

▸ 3.1 Installation targets

A package reaches a device only when that device is listed as an installation target. This is separate from the device being managed — registration in Lab 04 established management, and it did not assign any policy.

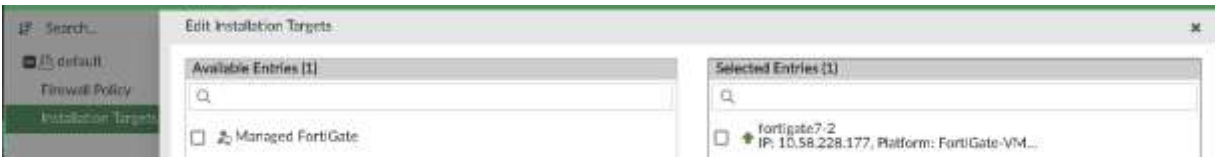


Figure 8 — Edit Installation Targets — fortigate7-2 selected for the default package.

▸ 3.2 Validation catches the object

The wizard validates before it installs, and this is where the zero mask surfaces.



Figure 9 — Install Wizard step 3 — Policy and Object Validation reporting LAN-Net set to 0.0.0.0/0.

The message is explicit: *Firewall address LAN-Net has IP set to 0.0.0.0/0, which means all IP addresses*. Interface Validation and Ready to Install are both green alongside it, so the wizard will proceed if told to.

A validation error is not a block

The red cross on Policy and Object Validation does not stop the install — the Install button stays active and the wizard continues.

This is the point of the step: FortiManager reports what it thinks is wrong and leaves the decision to the administrator. Reading the validation output is as important as reading the install preview in Lab 05.

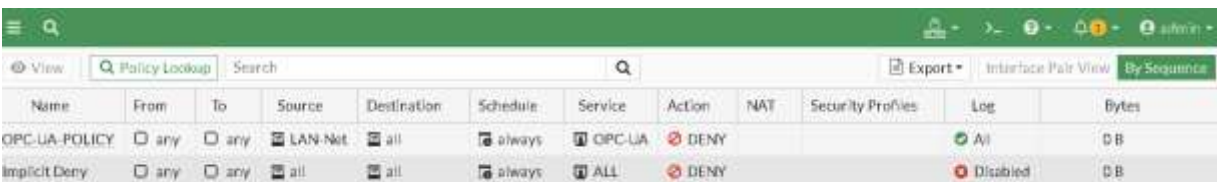
3.3 The device before and after

The FortiGate holds no policies at this point, only the implicit deny.



Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
Implicit									

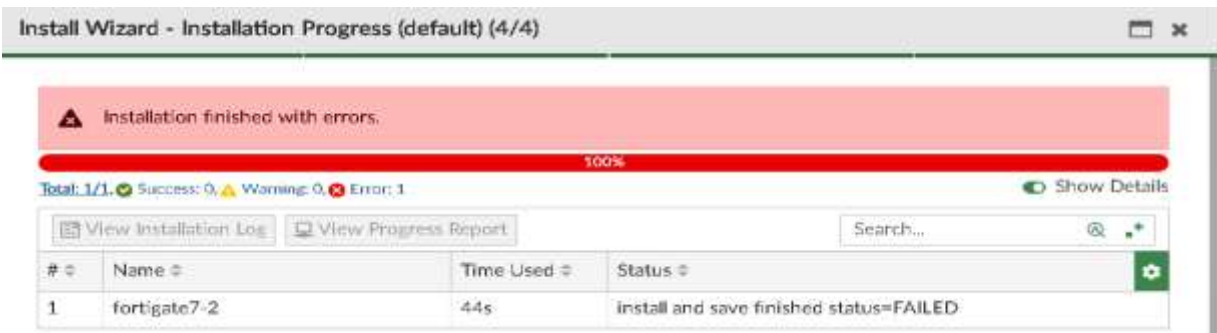
Figure 10 — FortiGate firewall policy list before the install.



Name	From	To	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
OPC-UA-POLICY	any	any	LAN-Net	all	always	OPC-UA	DENY			All	0 B
Implicit Deny	any	any	all	all	always	ALL	DENY			Disabled	0 B

Figure 11 — The same list after the install, with OPC-UA-POLICY present above the implicit deny.

3.4 The install reports failure



#	Name	Time Used	Status
1	fortigate7-2	44s	install and save finished status=FAILED

Figure 12 — Installation finished with errors — install and save finished status=FAILED.



Figure 13 — Device configuration status showing Conflict.

Observation

The policy is visible on the FortiGate, yet the install reported FAILED and the device moved to Conflict rather than Synchronized.

Reading

The policy content reached the device while the overall install-and-save transaction did not complete. The device and the database no longer agree on the full configuration, which is what Conflict means — distinct from the Modified states in Lab 05, where one copy is simply ahead of the other.

04. Resolving the Conflict

▸ 4.1 Correcting the object

The fix is the mask that was wrong in section 1.2.

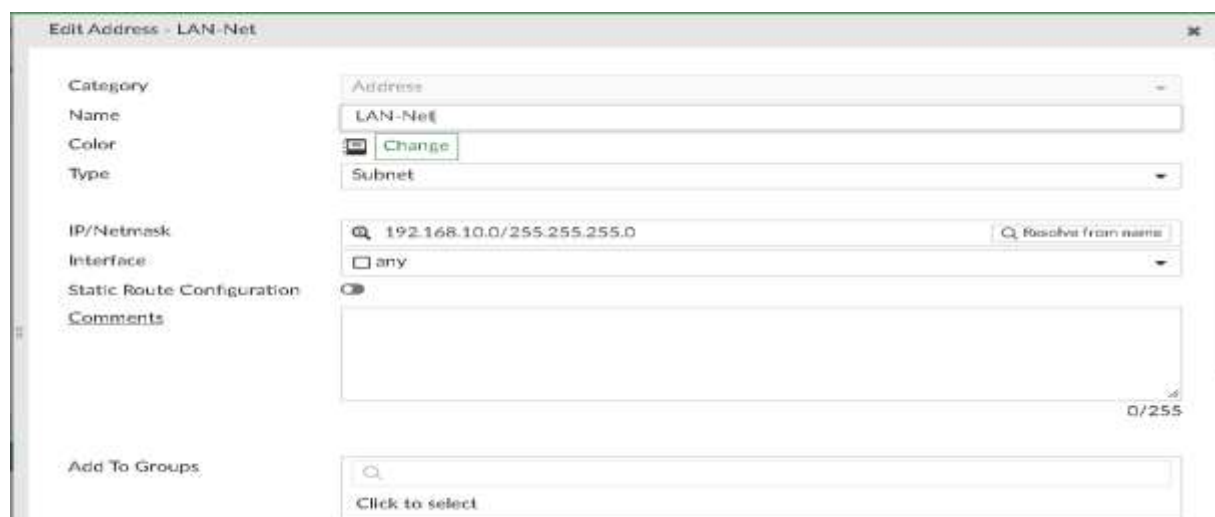


Figure 14 — LAN-Net edited to 192.168.10.0/255.255.255.0.

▸ 4.2 Re-validating

The wizard now reports Policy and Object Validation green alongside the other two checks.



Figure 15 — Install Wizard validation with no errors.

▸ 4.3 The revision diff

FortiManager compares the stored revision against the current device database as a line-by-line diff.

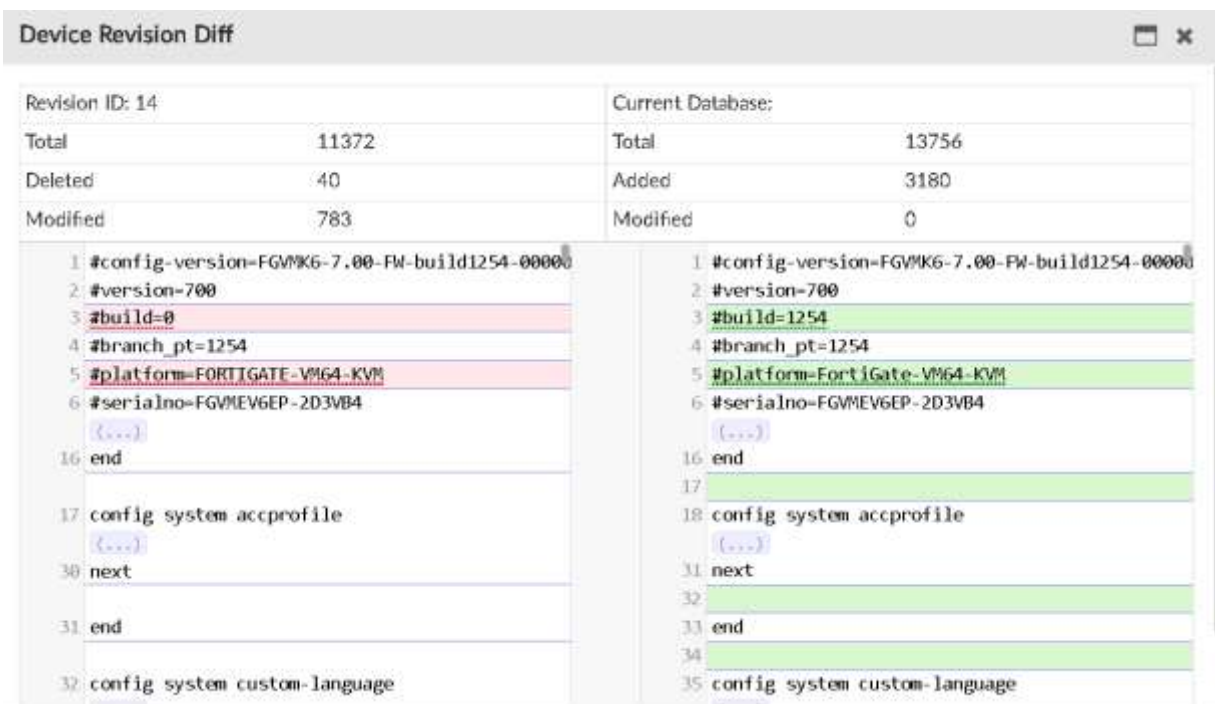


Figure 16 — Device Revision Diff — revision 14 against the current database.

Observation	The diff reports 3,180 lines added and 783 modified, far more than a policy change and a mask correction would produce. The header lines also differ: build=0 against build=1254, and FORTIGATE-VM64-KVM against FortiGate-VM64-KVM.
Reading	A revision recording build=0 and an uppercase platform string is characteristic of a capture taken while the device database was incomplete — consistent with revision 14 having been written around the failed install rather than from a healthy retrieve.
Action	Worth comparing against a revision taken before the failed install. If the earlier revisions also show build=0, the cause is the capture; if only revision 14 does, the failed install is what produced it.

05. Per-Device Mapping

An ADOM object has one definition shared by every device. Per-device mapping is the exception mechanism: the object keeps its name and its references, and one named device resolves it to a different value.

This is what makes a single policy package installable across sites whose networks differ. The policy says LAN-Net; each firewall receives its own subnet.

▸ 5.1 Creating the mapping

The screenshot shows the 'Per-Device Mapping' configuration window. It includes a dropdown menu for 'Mapped Device' with 'fortigate7-2' selected. Below this, there are fields for 'Map to Address' (containing '172.168.10.0/255.255.255.0'), 'Interface' (with a 'Click to select' button), and a 'Static Route Configuration' toggle switch. At the bottom is a 'Comments' text area. The window has a title bar 'Per-Device Mapping' and a close button. The bottom right corner indicates '0/255'.

Figure 17 — Per-Device Mapping — fortigate7-2 mapped to 172.168.10.0/255.255.255.0.



Figure 18 — The mapping listed against the object.

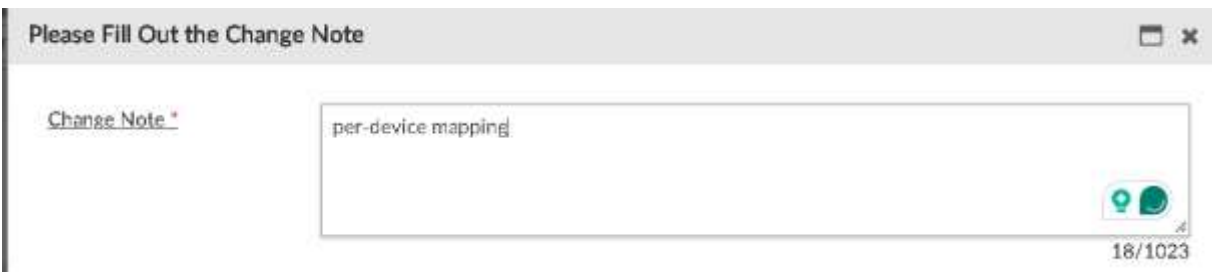


Figure 19 — Change note recorded for the mapping.

► 5.2 The effect on the device

Before the mapping is installed, the FortiGate holds the ADOM value.

IP Range/Subnet				
FABRIC_DEVICE	0.0.0.0/0		Address	0
FIREWALL_AUTH_PORTAL_ADDRESS	0.0.0.0/0		Address	0
LAN-Net	192.168.10.0/24		Address	2
SSLVPN_TUNNEL_ADDR1	10.212.134.200 - 10.212.134.210		Address	0
Site-LAN	10.58.228.0/24		Address	0
all	0.0.0.0/0		Address	0

Figure 20 — LAN-Net on the FortiGate as 192.168.10.0/24, the ADOM definition.

After the install it holds the mapped value, while the object in the ADOM is unchanged.

IP Range/Subnet				
FABRIC_DEVICE	0.0.0.0/0		Address	0
FIREWALL_AUTH_PORTAL_ADDRESS	0.0.0.0/0		Address	0
LAN-Net	172.168.10.0/24		Address	2
SSLVPN_TUNNEL_ADDR1	10.212.134.200 - 10.212.134.210		Address	0
Site-LAN	10.58.228.0/24		Address	0
all	0.0.0.0/0		Address	0

Figure 21 — LAN-Net on the FortiGate as 172.168.10.0/24, the per-device value.

One object, two values

The ADOM definition and the device value are now different, and both are correct. The policy still references a single object named LAN-Net.

This is also why the device value cannot be read as the source of truth for a mapped object. The ADOM holds the default; the mapping table holds the exceptions; the device shows whichever applied to it.

o6. Dynamic Objects

A dynamic address is a different mechanism that is easily confused with per-device mapping, and the distinction is an exam point.

The screenshot shows the 'Create New Address' window. The 'Category' is set to 'Address'. The 'Name' field contains 'dynamic-object1'. The 'Color' field has a 'Change' button. The 'Type' is set to 'Dynamic'. The 'Sub Type' is set to 'Fabric Connector Address'. The 'SDN Connector' is set to 'Click to select'. The 'Interface' is set to 'any'. The 'Static Route Configuration' is a toggle switch that is currently turned off. The 'Comments' field is empty, with a character count of '0/255'. At the bottom, there is an 'Add To Groups' section with a search bar and the text 'Click to select'.

Figure 22 — Create New Address with Type set to Dynamic and Sub Type Fabric Connector Address.

Mechanism	Where the value comes from	When it resolves
Per-device mapping	An administrator types a value for each named device in FortiManager.	At install time. FortiManager substitutes the mapped value for that device.
Dynamic address	An external source — an SDN or fabric connector, such as a cloud platform or a Fortinet fabric member.	On the device, continuously. Membership changes as the external source changes, with no install.

Per-device mapping is static content that varies by device. A dynamic object is a single definition whose content varies over time. Neither is a substitute for the other, and the metadata variables from Lab 06 are a third mechanism again — substitution inside script and template text rather than inside an object.